RMIT UNIVERSITY

CSIRO

RMIT Quantum Summer School 2026

Projects

DJ's & Grover's Algorithms

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The Deutsch–Jozsa's Problem

Given an **n-qubit** Oracle ("black box") function

We don't know exactly what is inside, but we know the promise that it is either :

- **constant** (i.e., always returns 0 or 1 whenever it's queried)

or

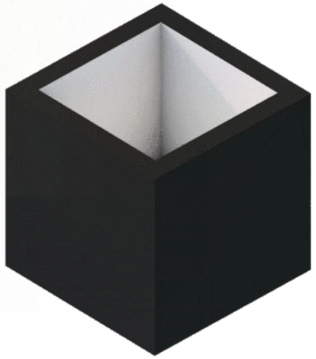
- **balanced** (i.e., 50% returns 0 and 50% returns 1).

→ Determine whether that oracle is constant or balanced

With classical solution, how many query do we need?

- ***The best case: 2 queries***
- ***The worst case: $2^n/2 + 1$ queries***

How about Quantum solution? ***→ 1 query!***



Oracle

- **Input**: n-bit
- **Output**: 0 or 1

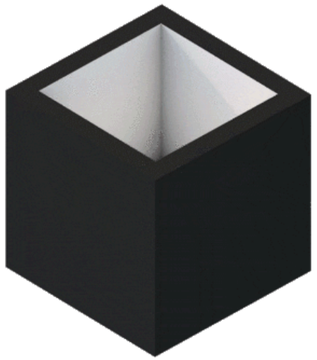
Project: Deutsch-Jozsa's Algorithm

Project Overview

The Deutsch-Jozsa algorithm (1992) was the first quantum algorithm to prove exponential speedup over classical computation. In this project, you will **implement** and **analyse** the **generalised Deutsch-Jozsa algorithm** for **n-qubit black-box** functions.

Deliverables

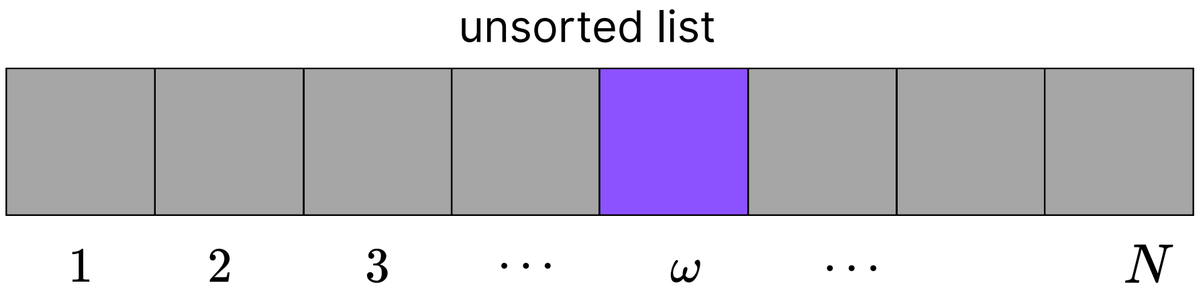
1. Multiple **oracle** functions demonstrating **constant** and **balanced** cases
2. Working implementation of the **n-qubit Deutsch-Jozsa algorithm**
3. Analysis of why the DJ algorithm works through the phase kickback mechanism.
4. Analysis of **circuit depth** and **gate complexity as n scales**
5. Comparison of quantum query complexity vs. classical approaches
6. Testing and validation on ideal and noisy quantum simulators



Oracle

- **Input**: n-bit
- **Output**: 0 or 1

The Unstructured Search Problem



Given an unsorted list with N = 2^n items

Exactly one *(or a few)* **is the "marked item"** ω

How can we find ω efficiently?

→ **an unstructured search problem:**

No ordering, no structure to exploit

(finding a needle in a haystack)

→ **Classical algorithm need on average** $N/2 = O(N)$ **times (the worst case: N)**

Cannot do better than linear search in the black-box (oracle) model

In quantum computing, we can use Grover's algorithm → need $O(\sqrt{N})$**times** → **Quadratic speed up**

Project: Grover’s Algorithm

Project Overview

In this project, you will implement and analyse **Grover's quantum search algorithm** for unstructured search in **n-qubit** systems. You will start with the classical case of searching for a **single marked item**, then extend your implementation to handle **k > 1 marked items**

Deliverables

1. Multiple **oracle** functions for different marked items,
2. Working implementation of the n-qubit single-item Grover’s search
3. An extended circuit implementation supporting **k = 2, 3, 4 marked items**
4. Analysis of the **optimal iteration count** vs. **k** relationship
5. Analysis showing how the Grover operator **amplifies target amplitudes**
6. Testing and validation on ideal and noisy quantum simulators